Rockborne Football League Database Documentation

This report presents the design and documentation of a relational database created for the Rockborne Football League Project.
The database was created to organize and manage data related to teams, players, matches, venues, and results. Initially, the raw data contained unstructured information such as match fixtures, scorers, and records stored in large, repetitive tables.

This documentation includes:

1. An explanation of the normalization process (from unnormalized data to 3NF).
2. An Entity–Relationship (ER) Diagram showing all entities and relationships.
3. A Data Dictionary describing each field, data type, and key.
4. A list of Database Assumptions used during the design process.

The goal of the Rockborne Football League Database is to create a clear and accurate data model that will help future analysis and reporting of player performance, team results, and match statistics.

Normalisation:

Firstly, I dealt with the Clubs names, participating groups and venues data. The problem was that there were no unique identifiers, the lists could not be linked to other tables (e.g., players or fixtures) and the data structure violates 1NF, as each list is not a structured relation with defined keys. (Refer to figure 1)

Scenario Raw Data

Club Names:

1. Data Masters
2. BI Gods
3. Vis Wizards
4. Data Cleaners

Participating groups:

1. Cohort 4
2. Cohort 5
3. Cohort 6
4. Cohort 7
5. Bench
6. Training Team
7. HR
8. Consultants

Venues:

1. Wimbledon 1
2. Wimbledon 2
3. Wimbledon 3

Figure 1

To achieve 1NF:

- Each item in the list was stored in its own record.
- Each table was assigned a primary key to uniquely identify every record.
- Data was organized into separate entity tables

Second Normal Form (2NF)

At this stage:

- Each table already depended fully on its primary key.
- No partial dependencies existed since each non-key attribute (the name) depends only on its unique ID.
- Therefore, all three tables already satisfied 2NF.

Third Normal Form (3NF)

To ensure 3NF:

- It was verified that there were no attribute depends on another non-key attribute.
- Each attribute (TeamName, GroupName, VenueName) depends only on its respective primary key.

This meant that all tables were satisfied 3NF and were therefore all normalised. Below is final look of one of the tables:

TeamID	TeamName
1	Data Masters
2	BI Gods
3	Vis Wizards
4	Data Cleaners

Figure 2(Team table normalised)

The fixtures data was given in a table format containing multiple weeks of match information, including the date, time, home team, away team, and venue. Although the data appeared to be organized, it was still in unnormalized form because it combined multiple sets of similar information into one large table with no unique identifier. Each week's fixtures were presented as separate sub-tables, which meant that there was no consistent structure to represent each match as an individual record. Additionally, there was no primary key to uniquely identify each fixture, and attributes such as team names and venue names were repeated multiple times across the dataset

Fixtures:

Week 1			
Date	**Home team**	**Away team**	**Venue**
01/10/2022 16:00	Data Masters	BI Gods	Wimbledon 1
01/10/2022 16:00	Vis Wizards	Data Cleaners	Wimbledon 2

Week 2			
Date	**Home team**	**Away team**	**Venue**
08/10/2022 16:00	Data Masters	Vis Wizards	Wimbledon 2
08/10/2022 16:00	BI Gods	Data Cleaners	Wimbledon 3

Week 3			
Date	**Home team**	**Away team**	**Venue**
22/10/2022 16:00	Data Masters	Data Cleaners	Wimbledon 3
22/10/2022 16:00	BI Gods	Vis Wizards	Wimbledon 2

Week 4			
Date	**Home team**	**Away team**	**Venue**
29/10/2022 16:00	BI Gods	Data Masters	Wimbledon 1
29/10/2022 16:00	Data Cleaners	Vis Wizards	Wimbledon 3

Week 5			
Date	**Home team**	**Away team**	**Venue**
05/11/2022 16:00	Vis Wizards	Data Masters	Wimbledon 3
05/11/2022 16:00	Data Cleaners	BI Gods	Wimbledon 1

Week 6			
Date	**Home team**	**Away team**	**Venue**
12/11/2022 16:00	Data Cleaners	Data Masters	Wimbledon 2
12/12/2022 16:00	Vis Wizards	BI Gods	Wimbledon 3

Figure 3

To bring the data into First Normal Form (1NF), repeating groups were removed. I added a new column, *Match ID*, to uniquely identify each fixture. Each field—such as date, time, home team, away team, and venue—contained only one value per record. This ensured that the table followed a structured and consistent format.

After achieving 1NF, the table was adjusted to meet Second Normal Form by removing partial dependencies. It was clear that attributes like *Home Team*, *Away Team*, and *Venue* contained data that repeated across several fixtures. These attributes were moved into their own tables each with a unique identifier. The Fixtures table then referenced these tables using foreign keys (*HomeTeamID*, *AwayTeamID*, and *VenueID*). This step eliminated redundancy and ensured that team and venue information was stored only once in the database.

Finally, to achieve Third Normal Form (3NF), the table was checked for transitive dependencies. There were no attributes that depended on non-key fields each non-key attribute depended directly on the table's primary key. At this stage, all redundant data had been removed, and the structure was fully normalized. The final *Fixtures* table contained the fields *MatchID*, *Date*, *Time*, *HomeTeamID*, *AwayTeamID*, and *VenueID*.

This fully normalized design allowed each fixture to be uniquely identified, reduced data repetition , and made it easy to establish relationships with other tables such as *Results* and *Events*.

MatchID	Date	Time	Home Team	Away Team	VenueID	IsFriendly
1	01/10/2022	16:00	1	2	1	No
2	01/10/2022	16:00	3	4	2	No
3	08/10/2022	16:00	1	3	2	No
4	08/10/2022	16:00	2	4	3	No
5	22/10/2022	16:00	1	4	3	No
6	22/10/2022	16:00	2	3	2	No
7	29/10/2022	16:00	2	1	1	No
8	29/10/2022	16:00	4	3	3	No
9	05/11/2022	16:00	3	1	3	No
10	05/11/2022	16:00	4	2	1	No
11	12/11/2022	16:00	4	1	2	No
12	12/12/2022	16:00	3	2	3	No
13	12/11/2023	16:00	4	1	1	Yes
14	12/11/2023	16:00	3	2	3	Yes

Figure 4

The results data was provided in a table format containing match outcomes, player actions, and event details, such as goals, yellow cards, and red cards. it was initially unnormalized because multiple sets of similar information were combined into single tables without unique identifiers. For example, goals, yellow cards, and red cards for each match were all listed together, and player names were repeated multiple times across different matches. There was no consistent structure to represent each match or event as an individual record, and no primary key existed to uniquely identify them.

Date - Time	Home Team * Player(s) scored ** Yellow card *** Red card		Away Team * Player(s) scored
01/10/2022 – 16:00	Data Masters * Alvin Ali (min 5) * Nathan Mcdowell (min 80)	2 – 0	BI Gods ** Isabelle Campos (min 10)
01/10/2022 – 16:00	Vis Wizards * Hayley Stevenson (min 11) ** Tanner Jenkins (min 30)	1 – 1	Data Cleaners * Dominik Cochran (min 10)
08/10/2022 – 16:00	Data Masters * Alvin Ali (min 2) * Alvin Ali (min 88) ** Aedan Petty (min 10) ** Jordin Christensen (min 70)	2 – 1	Vis Wizards * Hayley Stevenson (min 85)
08/10/2022 – 16:00	BI Gods * Jamya Dodson (min 20) * Tomas Powell (min 40)	2 – 1	Data Cleaners * Dominik Cochran (min 30) *** Paola Wilkerson (min 40)
22/10/2022 – 16:00	Data Masters	0 – 0	Data Cleaners

Figure 5

To bring the data into First Normal Form (1NF), repeating groups were removed and separate tables were created for match results and player events. A new column, ResultsID, was added to the Results table to uniquely identify each match outcome, while a separate

column, EventID, was added to the Events table to uniquely identify each event within a match. Each field—such as *Home Goals*, *Away Goals*, *PlayerID*, *EventType*, and *Minute*—contained only a single value per record. This ensured that both the Results and Events tables had a consistent structure.

After achieving 1NF, the dataset was brought into 2NF by removing partial dependencies. Match data were stored only in the Results table, while player-specific details like *PlayerID*, *EventType*, and *Minute* were stored separately in the Events table. The Events table was then linked to the Results table through a foreign key, *MatchID*. This separation reduced data redundancy and improved data integrity.

Finally, to achieve3NF, both tables were checked for transitive dependencies. All non-key attributes in the Results table depended solely on the primary key *ResultsID*, and all non-key attributes in the Events table depended solely on *EventID*. There were no dependencies between non-key attributes This means that the tables were now fully normalised.

ResultsID	MatchID	Home Goals	Away Goals
1	1	2	0
2	2	1	1
3	3	2	1
4	4	2	1
5	5	0	0
6	6	1	2
7	7	2	2
8	8	1	2
9	9	3	2
10	10	0	1
11	11	2	0
12	12	2	2
13	13		
14	14		

Figure 6

The player data was initially provided in a single table containing consultant IDs, player names, team names, and group names. the table was unnormalized because multiple attributes repeated across records. For example, team names and group names appeared multiple times for different players, and some players had overlapping group assignments. There was no unique identifier for each record beyond the Consultant ID, and player information was mixed with team and group information, which increased the potential for redundancy and inconsistencies.

Player names and their associated group and team:

Consultant ID	Name	Team Name	Group Name
200	Aedan Petty	Data Masters	Cohort 4
201	Aliza Santos	Data Masters	Cohort 6
201	Kaylynn Vaughan	Data Masters	Training Team
202	Arjun Bauer	Data Masters	Cohort 7
202	Lilian Huber	Data Masters	Cohort 5
203	Lizeth Roberts	Data Masters	Training Team
203	Nathan Mcdowell	Data Masters	HR
204	Alvin Ali	Data Masters	Consultants
204	Jordin Christensen	Data Masters	HR
205	Saul Blevins	Data Masters	HR
205	Carina Meza	BI Gods	Cohort 5
206	Isabelle Campos	BI Gods	Bench
206	Kyleigh Phelps	BI Gods	Cohort 4
207	Angela Wong	BI Gods	HR
207	Kole Rojas	BI Gods	HR
208	Martha Potts	BI Gods	Cohort 7
208	Tomas Powell	BI Gods	Training Team
209	Paxton Clarke	BI Gods	Bench
209	Jamya Dodson	BI Gods	Consultants
210	Georgia Clements	BI Gods	Consultants
210	Edwin Crawford	Vis Wizards	Cohort 5

To bring the data into First Normal Form (1NF), repeating groups were removed and each attribute would only contain a single value per record. A unique primary key, ConsultantID, was used to identify each player. Each player's team and group were linked to their record as separate attributes, ensuring a consistent and structured format.

Next, the data was adjusted to Second Normal Form (2NF) by removing partial dependencies. Team and group information, which repeated across multiple players, were moved into separate tables with their own unique identifiers (TeamID and GroupID). The main Players table then referenced these tables using foreign keys, eliminating redundancy and ensuring that team and group information was stored only once.

Finally, to achieve Third Normal Form (3NF), the tables were checked for transitive dependencies. All non-key attributes depended directly on the primary key (ConsultantID), and there were no attributes depending on other non-key fields. At this stage, the dataset was fully normalized, allowing each player to be uniquely identified, reducing data repetition, and making it easier to link player records with other tables such as Results and Events.

PlayerID	Consultant ID	Name	TeamID	GroupID
1	200	Aedan Petty	3	1
2	201	Aliza Santos	1	3
8	204	Alvin Ali	1	8
24	212	Anahi Reyes	3	5
14	207	Angela Wong	2	8
4	202	Arjun Bauer	1	4
39	219	Ashleigh Kaiser	4	6
26	213	Brody Gutierrez	3	4
11	205	Carina Meza	2	2
38	219	Cornelius Dods	4	2
35	217	Dominik Cochra	4	4
21	210	Edwin Crawforc	3	2
20	210	Georgia Clemer	2	8
27	213	Hayley Stevensc	3	3
12	206	Isabelle Campc	2	5

Figure 7

ER-Diagram

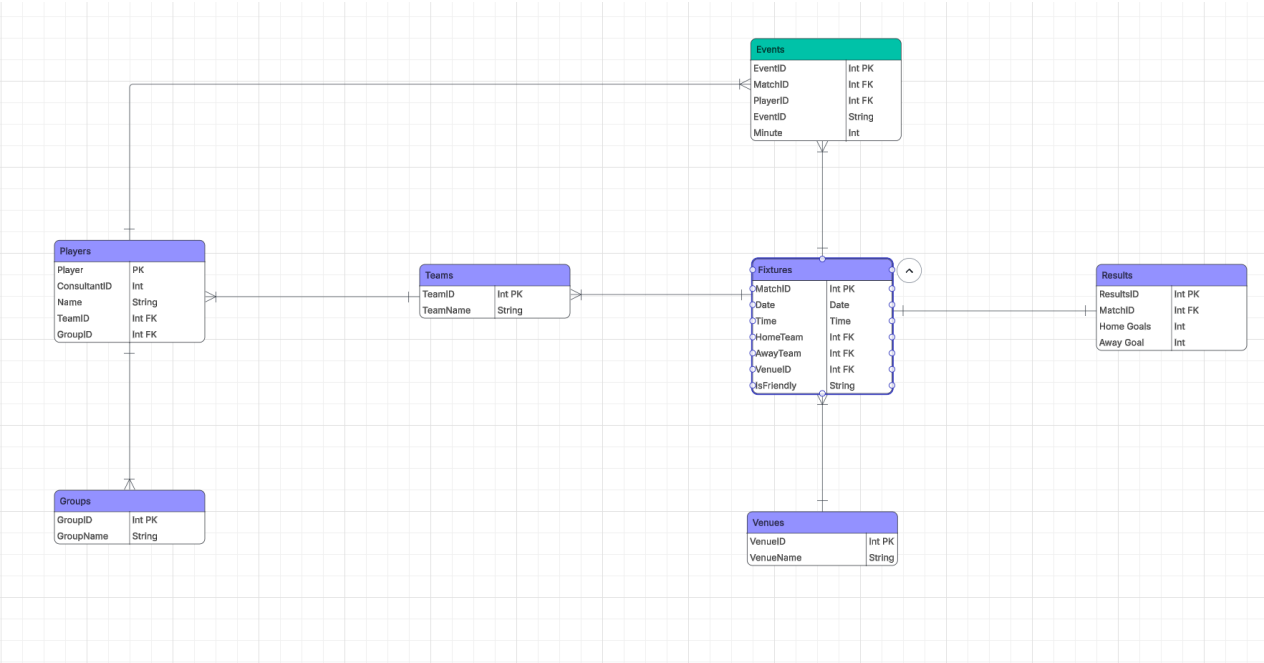
The Entity–Relationship (ER) Diagram for the Rockborne Football League database illustrates how all the main entities in the system are connected. The database design is based on the rules of the relational model. Each entity represents a distinct component of the football league, such as teams, players, matches, venues, results, and events. Relationships between these entities are clearly defined by using primary and foreign keys, which ensure referential integrity and minimize redundancy throughout the system.

The main entities identified are Teams, Groups, Players, Venues, Fixtures, Results, and Events. The Teams entity stores information about each football team, while the Groups entity represents the consultant groups from which players are drawn. The Players entity links individuals to both their team and their consultant group, using foreign keys (TeamID and GroupID) to establish these relationships. Each player has a unique PlayerID, which allows their performance records to be tracked across matches.

The Venues entity stores details of the football pitches where matches take place. The Fixtures entity records when and where each match occurs as well as which teams are playing. Each fixture has a unique MatchID and contains foreign keys that link to the Teams and Venues tables. The Results entity holds the scores for each fixture.

The Events entity records all individual player events that occur during a match, such as goals, yellow cards, and red cards. Each event is uniquely identified by EventID and contains

foreign keys to both MatchID and PlayerID. This structure allows detailed performance statistics to be stored without duplicating player or team data.



Relationships:

- Teams → Players = One-to-Many
- Groups → Players = One-to-Many
- Teams → Fixtures = One-to-Many
- Venues → Fixtures = One-to-Many
- Fixtures → Results = One-to-One
- Fixtures → Events = One-to-Many
- Players → Events = One-to-Many

Data dictionary

Table Name	Field Name	Data Type	Key	Description / Purpose
Teams	TeamID	INT	PK	Unique identifier for each football team.
	TeamName	VARCHAR(50)		Name of the football club.
Groups	GroupID	INT	PK	Unique identifier for each consultant group.
	GroupName	VARCHAR(50)		Name of the group (e.g., Cohort 4, HR).
Players	PlayerID	INT	PK	Unique identifier for each player.
	ConsultantID	INT		ID assigned to consultant/player.
	Name	VARCHAR(50)		Player's full name.
	TeamID	INT	FK	References the team the player belongs to.
	GroupID	INT	FK	References the group the player belongs to.
Venues	VenueID	INT	PK	Unique identifier for each venue.
	VenueName	VARCHAR(50)		Name or location of the football pitch (e.g., Wimbledon 1).
Fixtures	MatchID	INT	PK	Unique identifier for each match fixture.
	Date	DATE		Date of the match.
	Time	TIME		Kick-off time of the match.
	HomeTeamID	INT	FK	References the home team in the Teams table.
	AwayTeamID	INT	FK	References the away team in the Teams table.
	VenueID	INT	FK	References the venue where the match is played.
	IsFriendly	BOOLEAN		Indicates whether the match is part of the league or a friendly
Results	ResultID	INT	PK	Unique identifier for each match result.
	MatchID	INT	FK	Links the result to a specific fixture.
	HomeGoals	INT		Total goals scored by the home team.
	AwayGoals	INT		Total goals scored by the away team.
Events	EventID	INT	PK	Unique identifier for each individual match event.
	MatchID	INT	FK	References the fixture in which the event occurred.
	PlayerID	INT	FK	Identifies the player involved in the event.
	EventType	VARCHAR(20)		Type of event (Goal, Yellow Card, Red Card).
	Minute	INT		The minute during which the event occurred.

Data Assumptions

- The raw data was assumed to be unnormalised, containing repeated information about players, teams, matches, and events.
- It was assumed that each player, team, and venue should appear only once in the database to remove redundancy.
- Because some players shared the same Consultant ID, it was assumed that a new PlayerID field was needed to uniquely identify each player record.
- The Results table in the data mixed match details and event outcomes together. It was assumed these should be separated into two differnet entities
- The Score column (e.g., "2 – 1") was assumed to be the value, calculated from the total number of goals scored by each team.
- Blank fields (e.g., missing yellow or red cards) were assumed to mean no event occurred, not missing data.

- Where multiple home scorers were listed in one cell (e.g., “Alvin Ali; Nathan Mcdowell”), these were separated into multiple records